



INDIA.RUNS

Build what next India runs on

Project · HireFit | Team · Pakka

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Problem Statement · Rank the best 100 candidates from 100,000 profiles for a Senior AI Engineer role.
The system must remain offline, CPU-only, deterministic, and explainable.

Solution Overview

An evidence-first recruiter that reads careers and behaviour, ranks on proof, and explains every decision.

100K

candidate pool

Top-100

final shortlist

105.3 s

100K on 2 CPU

**Traceable
base + rank steps**

Aspect	Common public demos	HireFit
Matching	keywords or semantic similarity	evidence of shipped work (33 features)
Evaluation	often one score or a sample demo	career + skill + behaviour + logistics + integrity
Integrity	varies; frequently undocumented	review flags + hard gates; 0 detector flags in top-100
Explainability	similarity score or generated rationale	exact evidence-base contributions + documented rank steps
Reproducibility	full-pool artifact often absent	byte-identical full-pool output; frozen SHA-256

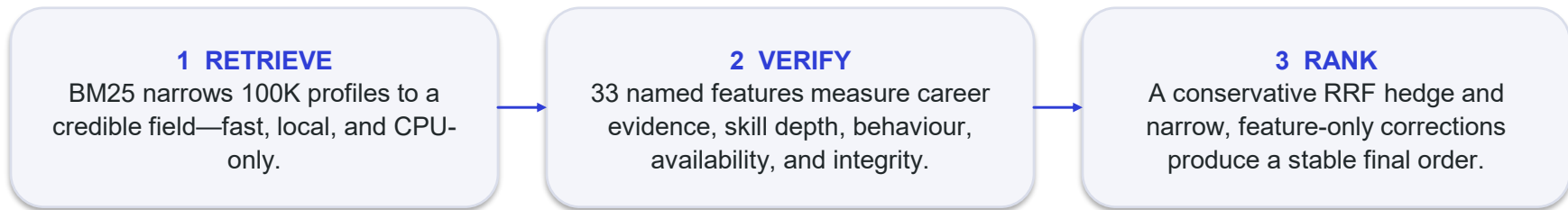
JD Understanding & Candidate Evaluation

The JD lists keywords; the role hires for evidence. We translate requirements into signals that survive scrutiny.

What the JD says	What the role actually hires for
5-9 yrs; embeddings, retrieval, ranking, NLP, PyTorch/TensorFlow; India locations	demonstrated evidence of shipping production retrieval/ranking systems
a list of AI skill keywords	production depth and verified application—not keyword count alone
a candidate profile	role & seniority alignment, availability & recruiter-responsiveness, and profile integrity

Ranking Methodology

Retrieve the credible field, score the evidence, then rank with a stable, inspectable sort.



Evidence base: 33 deterministic features computed from profile facts—not generated text.

Hard guardrails: integrity-anomaly and JD-disqualifier multipliers act before the final sort.

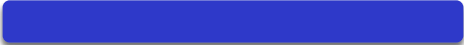
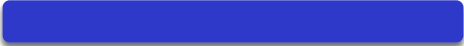
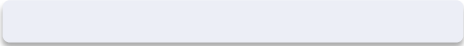
Stable release: two clean integrity backfills and two narrow tie-breaks survived holdout testing.

Grounded output: no hosted LLM or API at rank time; reasons use observed candidate facts only.

Explainability & Data Validation

A judge can trace the evidence score, the grounded reason, and every integrity decision.

Verified explanation & release checks

Evidence features		33
Top-100 reasons		100
Review signals		55
Flags shortlisted		0
Golden output		3d2d...
Faithfulness test		PASS

01 / EXACT ATTRIBUTION

Each feature's additive term is its analytic Shapley value. The terms reconstruct the evidence score exactly.

02 / GROUNDED REASONING

Candidate reasons use deterministic templates populated only from observed profile facts; no text generator runs at rank time.

03 / INTEGRITY REVIEW

55 profiles were flagged for review; 0 entered the top 100. The new three-way YoE check closes two previously missed contradictions without using candidate IDs.

End-to-End Workflow

One offline command turns 100,000 profiles into a validated, explained top-100.



ONE COMMAND `PYTHONHASHSEED=0 python rank.py --release --candidates candidates.jsonl --out submission.csv --workers 2`

PUBLISH GATE 105.3 s pipeline / 117.2 s wall • byte-identical SHA-256 3d2dbd8a... • membership, count, integrity, and hash checks before atomic publish.

System Architecture

Six CPU-only stages sit inside a fail-closed release envelope; optional ideas ship only when measurement supports them.



FAIL-CLOSED RELEASE Verifies input, model, backend, full-pool counts, integrity counts, and final output hash before atomic publish.

LIMITS Quality figures are development proxies for one role, not official hidden labels. Human review remains required.

FAILURE SAFETY A forced container OOM exits 137, preserves the prior output, and leaves no temporary publication file.

Results & Performance

The result is useful because quality, runtime, safety, and reproducibility can all be checked independently.

RANKING QUALITY

100,000

candidates scored

100

final shortlist

0

detector flags in top-100

278

tests passed

RUNTIME & COMPUTE

105.3 s

100K pipeline

117.2 s

wall @ 2 CPU

~4.2 GB

peak memory

< 300 s

budget, offline

✓ Deterministic

✓ SHA-256
3d2dbd8a

✓ CPU-only / offline

✓ Validator passed

✓ 0 detector
flags✓ No hidden-score
claim

Technologies Used

Every dependency earns its place: fast on CPU, offline at rank time, deterministic, and easy to reproduce.

01 / RANKING CORE

Python 3.11 + NumPy

Deterministic CPU numerics

bm25s + fallback

Fast lexical retrieval over 100K

130 KB NumPy model

No GPU or PyTorch dependency

02 / LIVE DELIVERY

FastAPI + static UI

Recruiter decision room on Render

Gradio

Inspectable Hugging Face sandbox

Two live surfaces

One consistent ranking story

03 / RELEASE PROOF

Pinned environment

Docker digest + dependency hashes

278 checks

All collected checks pass

Golden release

Output hash + 2K behaviour gate

A deliberately small stack keeps every run fast, reproducible, and easy for judges to inspect.

Submission Assets

A complete judge path: experience the product, inspect the proof, reproduce the output, then present the result.

01 / TRY IT

[Render decision room ↗](#)

Review the ranked shortlist and evidence.

[Hugging Face sandbox ↗](#)

Run the inspectable demo surface.

02 / VERIFY IT

[Open source repository ↗](#)

Read the code, tests, and README.

Judge proof

docs/JUDGE_PROOF.md

Reproduce

```
python rank.py --release
```

03 / DOWNLOAD

submission.csv

Predefined top-100 format

ranked_top100.xlsx

Grounded reasons for review

Validator passed

SHA-256 3d2dbd8a...

04 / PRESENT

Live walk-through

Reproduce → validate → rank
→ explain

Presenter guide

docs/DEMO_SCRIPT.md

Close with proof

Live links + deterministic
result



INDIA.RUNS

Build what next India runs on

THANK YOU

VERIFY IT YOURSELF github.com/bansalbhunesh/redrob-hirefit-ranker
LIVE redrob-hirefit-ranker.onrender.com • huggingface.co/spaces/bansal1234/Hirefit